

## PART TWO

# Wave Optics

### 11.5 TRANSVERSE WAVES

All light waves are classified as *transverse waves*. Transverse waves are those in which each small part of the wave vibrates along a line perpendicular to the direction of propagation and all parts are vibrating in the same plane. A wave machine for demonstrating transverse waves is shown in Fig. 11G. When the handle *H* is turned clockwise the small white balls at the top of the vertical rods move up and down with SHM.

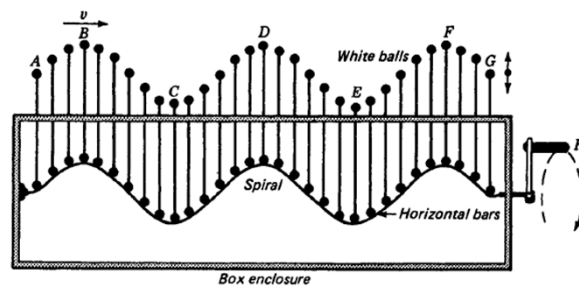


FIGURE 11G  
Machine for demonstrating transverse waves.

## 11.6 SINE WAVES

The simplest kind of wave train is that for which the motions of all points along the wave have displacements  $y$  given by the *sine* or *cosine* of some uniformly increasing function. This in effect describes what we have called SHM.

Consider transverse waves in which the motions of all parts are perpendicular to the direction of propagation. The displacement  $y$  of any point on the wave is then given by

$$y = a \sin \frac{2\pi x}{\lambda} \quad (11r)$$

A graph of this equation is shown in Fig. 11I, and the significance of the constants  $a$  and  $\lambda$  is clear. To make the wave move to the right with a velocity  $v$ , we introduce the time  $t$  as follows:

$$y = a \sin \frac{2\pi}{\lambda} (x - vt) \quad (11s)$$

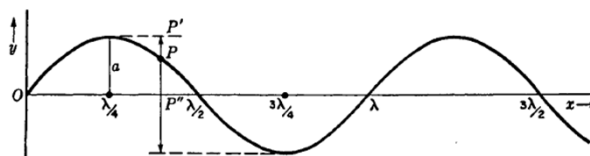


FIGURE 11I  
Contour of a sine wave at time  $t = 0$ .

## 11.7 PHASE ANGLES

In wave motion the instantaneous displacement and direction of propagation are described by specifying the position of the graph point on the circle of reference (see Fig. 11J). The angle  $\theta$ , measured counterclockwise from the  $+x$  axis, specifying the position is called the *phase angle*. As an example consider a point moving up and down along the  $y$  axis, as shown in Fig. 11J. The position of the mass point  $P$  is given by the projection of the graph point  $p$  on the  $y$  axis. From the right triangle  $PpC$  on the diagram

$$y = a \sin \theta \quad (11v)$$

With the graph point moving at constant speed  $v$ , the angular speed  $\omega$  is constant, and we can write for any angle  $\theta$

$$\theta = \omega t$$

Substitution in Eq. (11v) gives

$$y = a \sin \omega t \quad (11w)$$

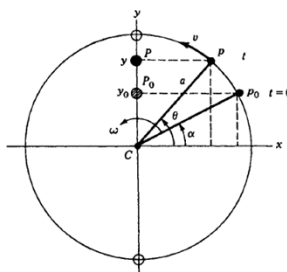


FIGURE 11J  
Simple harmonic motion along the  $y$  axis, showing the circle of reference, the initial phase angle  $\alpha$ , the angular speed  $\omega$ , and the point  $P_0$  at time  $t = 0$ .

At time  $t = 0$  the graph point is at  $+p_0$  and the mass point is at  $P_0$ . At some later time  $t$  when the mass point is at  $P$ , the graph point is at  $p$  and we must modify Eq. (11w) by adding the angle  $\alpha$  as follows:

$$y = a \sin(\omega t + \alpha) \quad (11x)$$

The angle  $\alpha$  is a constant and is called the *initial phase angle*.

**EXAMPLE 2** A given point is vibrating with SHM with a period of 5.0 s and an amplitude of 3.0 cm. If the initial phase angle is  $\pi/3$  rad, ( $60^\circ$ ), find (a) the initial displacement and (b) the displacement after 12.0 s. (c) Make a graph.

**SOLUTION** (a) Since the graph point makes one revolution in 5.0 s, the angular speed  $\omega$  is  $2\pi$  rad in 5.0 s, or  $2\pi/5$  rad/s [see Eq. (11f)]. At the time  $t = 0$ , direct substitution in Eq. (11x) gives

$$y = 3 \sin\left(\frac{2\pi}{5} 0 + \frac{\pi}{3}\right)$$

(b) After 12.0 s, substitution in Eq. (11x) gives

$$y = 3 \sin\left(\frac{2\pi}{5} 12 + \frac{\pi}{3}\right) = 3 \sin\left(4.8\pi + \frac{\pi}{3}\right)$$

The total phase angle of  $4.8\pi + \pi/3$  is equivalent to  $864^\circ + 60^\circ$ , or  $924^\circ$

## 11.8 PHASE VELOCITY AND WAVE VELOCITY

The discussion given in connection with Fig. 11K may be summed up by saying that a wave constitutes the progression of a condition of constant phase. This condition might be, for instance, the crest of the wave, where the phase is such as to yield the maximum upward displacement. The speed with which a crest moves along is usually called the wave velocity, although the more specific term phase velocity is sometimes used.

$$\text{the wave velocity} \quad v = \frac{dx}{dt} = \frac{\omega}{k} \quad (11z)$$

The ratio  $\omega/k$  for a given kind of wave depends on the physical properties of the medium in which the waves travel and also, in general, on the frequency  $\omega$  itself.

### 11.9 AMPLITUDE AND INTENSITY

Waves transport energy, and the amount of it that flows per second across unit area perpendicular to the direction of travel is called the *intensity* of the wave. If the wave flows continuously with the velocity  $v$ , there is a definite energy density, or total energy per unit volume. All the energy contained in a column of the medium of unit cross section and of length  $v$  will pass through the unit of area in 1 s. Thus the intensity is given by the product of  $v$  and the energy density. Either the energy density or the intensity is proportional to the square of the amplitude and to the square of the frequency.

In spherical waves, the intensity decreases as the inverse square of the distance from the source. This law follows directly from the fact that, provided there is no conversion of the energy into other forms, the same amount must pass through any sphere with the source as its center. Since the area of a sphere increases as the square of its radius, the energy per unit area at a distance  $r$  from the source, or the intensity, will vary as  $1/r^2$ . The amplitude must then vary as  $1/r$ , and one may write the equation of a spherical wave as

$$y = \frac{a}{r} \sin(\omega t - kr) \quad (11zc)$$

Here  $a$  means the amplitude at unit distance from the source.

$$I_x = I_0 e^{-ax} \quad (11zd)$$

This law, which has been attributed to both Bouguer\* and Lambert,† we shall refer to as the *exponential law of absorption*.

### 11.10 FREQUENCY AND WAVELENGTH

Any wave motion is generated by some sort of vibrating source, and the frequency of the waves is equal to that of the source. The wavelength in a given medium is then determined by the velocity in that medium and by Eq. (11t) is obtained by dividing the velocity by the frequency. Passage from one medium to another causes a change in the wavelength in the same proportion as it does in the velocity, since the frequency is not altered.

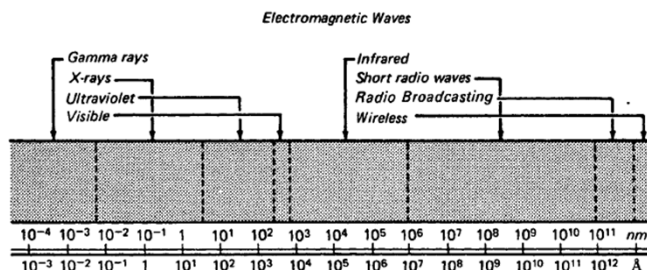


FIGURE 11N  
Scale of wavelengths for the range of known electromagnetic waves.

## 12

## THE SUPERPOSITION OF WAVES

When two sets of waves are made to cross each other, e.g., the waves created by dropping two stones simultaneously in a quiet pool, interesting and complicated effects are observed. In the region of crossing there are places where the disturbance is practically zero and others where it is greater than that given by either wave alone. A very simple law can be used to explain these effects, which states that the resultant displacement of any point is merely the sum of the displacements due to each wave separately. This is known as the *principle of superposition* and was first clearly stated by Young\* in 1802.

## 12.1 ADDITION OF SIMPLE HARMONIC MOTIONS ALONG THE SAME LINE

we can write the separate displacements as follows:

$$\begin{aligned} y_1 &= a_1 \sin(\omega t - \alpha_1) \\ y_2 &= a_2 \sin(\omega t - \alpha_2) \end{aligned} \quad (12a)$$

Note that  $\omega$  is the same for both waves, since we have assumed them to be of the same frequency. According to the principle of superposition, the resultant displacement  $y$  is merely the sum of  $y_1$  and  $y_2$ , and we have

$$y = a_1 \sin(\omega t - \alpha_1) + a_2 \sin(\omega t - \alpha_2)$$

When the expression for the sine of the difference of two angles is used, this can be written

$$\begin{aligned} y &= a_1 \sin \omega t \cos \alpha_1 - a_1 \cos \omega t \sin \alpha_1 + a_2 \sin \omega t \cos \alpha_2 - a_2 \cos \omega t \sin \alpha_2 \\ &= (a_1 \cos \alpha_1 + a_2 \cos \alpha_2) \sin \omega t - (a_1 \sin \alpha_1 + a_2 \sin \alpha_2) \cos \omega t \end{aligned} \quad (12b)$$

Now since the  $a$ 's and  $\alpha$ 's are constants, we are justified in setting

$$\begin{aligned} a_1 \cos \alpha_1 + a_2 \cos \alpha_2 &= A \cos \theta \\ a_1 \sin \alpha_1 + a_2 \sin \alpha_2 &= A \sin \theta \end{aligned} \quad (12c)$$

provided that constant values of  $A$  and  $\theta$  can be found which satisfy these equations. Squaring and adding Eqs. (12c), we have

$$\begin{aligned} A^2(\cos^2 \theta + \sin^2 \theta) &= a_1^2(\cos^2 \alpha_1 + \sin^2 \alpha_1) + a_2^2(\cos^2 \alpha_2 + \sin^2 \alpha_2) \\ &\quad + 2a_1a_2(\cos \alpha_1 \cos \alpha_2 + \sin \alpha_1 \sin \alpha_2) \\ \text{or} \quad A^2 &= a_1^2 + a_2^2 + 2a_1a_2 \cos(\alpha_1 - \alpha_2) \end{aligned} \quad (12d)$$

Dividing the lower equation (12c) by the upper one, we obtain

$$\tan \theta = \frac{a_1 \sin \alpha_1 + a_2 \sin \alpha_2}{a_1 \cos \alpha_1 + a_2 \cos \alpha_2} \quad (12e)$$

Equation (12d) and (12e) show that values of  $A$  and  $\theta$  exist which satisfy Eqs. (12c), and we can rewrite Eq. (12b), substituting the right-hand members of Eq. (12c). This gives

$$y = A \cos \theta \sin \omega t - A \sin \theta \cos \omega t$$

which has the form of the sine of the difference of two angles and can be expressed as

$$y = A \sin(\omega t - \theta) \quad (12f)$$

This equation is the same as either of our original equations for the separate simple harmonic motions but contains a new amplitude  $A$  and a new phase constant  $\theta$ .

The resultant amplitude  $A$  depends, according to Eq. (12d), upon the amplitudes  $a_1$  and  $a_2$  of the component motions and upon their difference of phase  $\delta = \alpha_1 - \alpha_2$ . When we bring together two beams of light, as is done in the Michelson interferometer (Sec. 13.8), the intensity of the light at any point will be proportional to the square of the resultant amplitude. By Eq. (12d) we have, in the case where  $a_1 = a_2$ ,

$$I \approx A^2 = 2a^2(1 + \cos \delta) = 4a^2 \cos^2 \frac{\delta}{2} \quad (12g)$$

If the phase difference is such that  $\delta = 0, 2\pi, 4\pi, \dots$ , this gives  $4a^2$ , or 4 times the intensity of either beam. If  $\delta = \pi, 3\pi, 5\pi, \dots$ , the intensity is zero. For intermediate values, the intensity varies between these limits according to the square of the cosine.

These modifications of intensity obtained by combining waves are referred to as *interference* effects, and we shall discuss in the next chapter several ways in which they can be brought about and used experimentally.

## 12.2 VECTOR ADDITION OF AMPLITUDES

A very simple geometrical construction can be used to find the resultant amplitude and phase constant of the combined motion in the above case of two simple harmonic motions along the same line. If we represent the amplitudes  $a_1$  and  $a_2$  by vectors making angles  $\alpha_1$  and  $\alpha_2$  with the  $x$  axis,\* as in Fig. 12A(a), the resultant amplitude  $A$  is the vector sum of  $a_1$  and  $a_2$  and makes an angle  $\theta$  with that axis. To prove this proposition, we first note from Fig. 12A(b) that in the triangle formed by  $a_1$ ,  $a_2$ , and  $A$  the law of cosines gives

$$A^2 = a_1^2 + a_2^2 - 2a_1a_2 \cos [\pi - (\alpha_1 - \alpha_2)] \quad (12h)$$

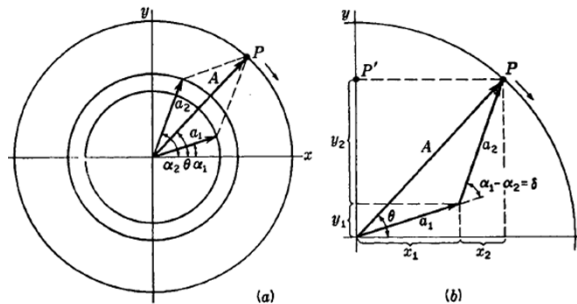


FIGURE 12A  
Graphical composition of two waves of the same frequency, but different amplitude and phase.

### 12.3 SUPERPOSITION OF TWO WAVE TRAINS OF THE SAME FREQUENCY

From the preceding section we can conclude directly that the result of superimposing two trains of sine waves of the same frequency and traveling along the same line is to produce another sine wave of that frequency but having a new amplitude which is determined for given values of  $a_1$  and  $a_2$  by the phase difference  $\delta$  between the motions imparted to any particle by the two waves. As an example, let us find the resultant wave produced by two waves of equal frequency and amplitude traveling in the same direction  $+x$ , but with one a distance  $\Delta$  ahead of the other. The equations of the two waves, in the form of Eq. (11y), will be

$$y_1 = a \sin (\omega t - kx) \quad (12i)$$

$$y_2 = a \sin [\omega t - k(x + \Delta)] \quad (12j)$$

By the principle of superposition, the resultant displacement is the sum of the separate ones, so that

$$y = y_1 + y_2 = a\{\sin (\omega t - kx) + \sin [\omega t - k(x + \Delta)]\}$$

Applying the trigonometric formula

$$\sin A + \sin B = 2 \sin \frac{1}{2}(A + B) \cos \frac{1}{2}(A - B) \quad (12k)$$

we find

$$y = 2a \cos \frac{k\Delta}{2} \sin \left[ \omega t - k \left( x + \frac{\Delta}{2} \right) \right] \quad (12l)$$

This corresponds to a new wave of the same frequency but with the amplitude  $2a \cos (k\Delta/2) = 2a \cos (\pi\Delta/\lambda)$ . When  $\Delta$  is a small fraction of a wavelength, this amplitude will be nearly  $2a$ , while if  $\Delta$  is in the neighborhood of  $\frac{1}{2}\lambda$ , it will be practically zero.

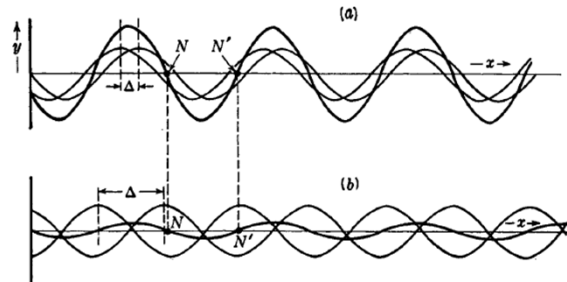


FIGURE 12C  
Superposition of two wave trains (a) almost in phase and (b) almost 180° out of phase.